GR Chapter 2: Off-center LTB sketch for Migkas anisotropy

C. A.

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Abstract

This note is a first-shot general-relativistic (GR) companion to the Newtonian Sedov–Taylor + Center-of-Expansion (ST+CoE) synthesis. The backbone is off-center dust Lemaître–Tolman–Bondi (LTB) geometry—optionally with Λ—chosen as the simplest metric promotion of the Newtonian Gauss enclosed-mass picture with radially decreasing density. We state the LTB metric, encode a centrally concentrated toy $M(r)$, place an off-center observer, and map a direction-dependent expansion rate to the Migkas et al. (2020) sheet convention $A \propto H_{\rm obs}/H$. A numerical toy (FLRW-shell $H(r)$, not full null geodesics) yields affine $R^2 \approx 0.43$ against digitised Migkas $A(\ell)$ after a sky-axis phase; Amp$\times$ sin and 2ℓ remain diagnostic residuals, not free GR parameters. The same toy, with a linear Hubble-gradient CMB dipole estimate, places a horizon-scale offset ($d \sim 0.5\,D$ with $D \sim c/H_0$) many orders of magnitude above the observed CMB dipole, consistent with the Alnes–Kodama $\sim 15\,$Mpc ceiling class for spherical off-center models. This is not a finished GR cosmology; it is an honest promotion of the Newtonian sky template into metric language and an explicit CMB kill-shot ledger.

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1 Purpose

Chapter 1 of this program [Migkas et al., 2020] (Newtonian synthesis, companion paper in this workspace) produced a phenomenological sky template: Sedov–Taylor homology about a real Center of Expansion (CoE), off-center observer, Gauss gravity from decreasing $\rho(r)$ (not a uniform ball $g \propto -r$), and a sheet map

$$A \;\propto\; H_{\rm obs}/H \tag{1}$$

aimed at Migkas et al. Fig. 6 A versus Galactic longitude. Geometry alone reached $R^2 \approx 0.543$; Amp$\times$ sin (Amp$= -0.06$, $\phi \approx 0.20\,$rad) raised $R^2 \approx 0.746$; a multiplicative 2ℓ layer reached $R^2 \approx 0.875$. Those last two layers are diagnostic, not first principles.

What this Chapter 2 does.

1. Promote the Newtonian ST+Gauss sky template to a **metric**: dust LTB (optional Λ), with decreasing $\rho(r)$ encoded in $M(r)$ / curvature / bang time.
2. State the off-center observer geometry (d/D, angles $\to$ Galactic longitude schematic).
3. Write an observable pipeline from (approximate) null-ray / local-H quantities to $H_{\rm obs}(\hat{n})$ and sheet A.
4. Run a **toy** numerical calculation with honest approximations, report real R^2 against digitised Migkas data, and draw a CMB dipole versus offset sketch in the Alnes–Kodama class.
5. Record what would be needed for a beyond-horizon CoE to survive the CMB, and what upgrades (Bertschinger, Smoller–Temple) come later—and what we are *not* doing (Bianchi).

What this Chapter 2 does not claim. It does not claim a finished GR cosmology, a fit competitive with ΛCDM, a solution of the CMB dipole problem, or that Amp$\times$ sin / 2ℓ emerge from the Einstein equation. Full LTB null-geodesic luminosity distances and exact temperature maps are deferred; the numerics here are scaffolding labeled as such.

2 LTB metric and decreasing density

2.1 Standard form

The Lemaître–Tolman–Bondi metric for a dust (pressureless) spherically symmetric spacetime is [Lemaître, 1933, Tolman, 1934, Bondi, 1947]

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \frac{\big(R'(t,r)\big)^2}{1-k(r)}\,\mathrm{d}r^2 + R(t,r)^2\,\mathrm{d}\Omega^2, \tag{2}$$

where $R(t,r)$ is the areal radius, prime denotes $\partial/\partial r$, and $k(r) < 1$ is the curvature function. The Einstein equation with dust density ρ and optional cosmological constant Λ integrates to

$$\dot{R}^2 = \frac{2GM(r)}{R} - k(r)\,R^2 + \frac{\Lambda}{3}R^2, \tag{3}$$

$$4\pi\rho = \frac{M'(r)}{R^2 R'}, \tag{4}$$

with $M(r)$ the active gravitational (Misner–Sharp) mass function. Bang time $t_B(r)$ enters through the integral of (3). Gauge freedom in r is often fixed by $R(t_0, r) = r$ at the present time t_0.

Dust LTB without Λ is the backbone used below; adding Λ does not change the geometric off-center story at the level of this sketch (cf. LTB+Λ void literature [Alnes and Amarzguioui, 2006, Blomqvist and Mörtsell, 2010]).

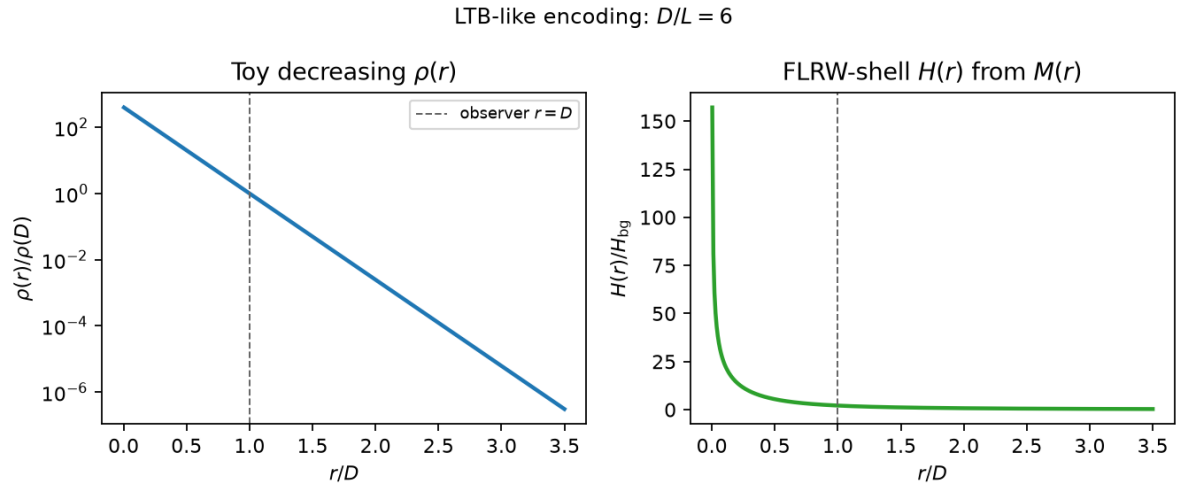


Figure 1: Toy decreasing $\rho(r)$ from (6) and the corresponding FLRW-shell $H(r)$ from (7), with observer at $r = D$ and $D/L = 6$. Centrally concentrated $M(r)$ produces higher H near the CoE. This figure is the density/curvature encoding step only—not yet an off-center sky map.

2.2 Encoding decreasing $\rho(r)$

Newtonian Chapter 1 insisted on decreasing density toward the outskirts so that Gauss gravity is $g = -GM(< r)/r^2$ with nontrivial radial structure—not a uniform ball. In LTB the same physics sits in $M(r)$ (and/or $k(r)$, $t_B(r)$).

Toy profile. We take a centrally concentrated enclosed mass

$$M(r) = M_\infty \Big[1 - e^{-x}(1 + x)\Big], \qquad x \equiv r/L, \tag{5}$$

so that

$$\rho(r) = \frac{M_\infty}{4\pi L^3}\, e^{-r/L} \tag{6}$$

decreases outward on scale L. This is an explicit toy, not a fit to large-scale structure. The Newtonian ballpark $D/L \sim$ few is retained ($D/L = 6$ in the numerics).

FLRW-shell Hubble function (approximation). Rather than integrating the full LTB characteristic equation for every ray, the toy replaces the metric evolution by a stack of dust FLRW shells,

$$H(r)^2 \equiv \left(\frac{\dot R}{R}\right)^2 \sim \frac{8\pi G}{3}\,\langle\rho\rangle(r) = \frac{2GM(r)}{R^3}, \tag{7}$$

matched to a background $H_{\rm bg}$ at large $r_{\rm bg}$. This is **not** a full LTB solution; it is the most direct metric-language translation of Newtonian enclosed-mass intuition. Captions and Section 5 mark the approximation.

3 Off-center observer geometry

Place the CoE at the coordinate origin. The observer sits at areal / radial coordinate distance D from the CoE. A unit direction $\hat n$ on the observer's sky makes an angle φ with the CoE–observer axis: $\varphi = 0$ looks anti-center (away from the CoE), $\varphi = \pi$ looks toward the CoE. In a planar toy,

$$\hat n = (\cos\varphi, \sin\varphi), \qquad \mu \equiv \hat n \cdot \hat D = \cos\varphi. \tag{8}$$

Sources at fixed observer-centric distance s (survey-depth stand-in) have CoE-radius

$$r(\varphi) = \sqrt{D^2 + s^2 + 2Ds\cos\varphi}. \tag{9}$$

The dimensionless ratios of interest match Chapter 1: d/D (here s/D on the source circle; observer offset $/D = 1$ in units), D/L, and mass contrast through $M(r)$.

Galactic longitude schematic. Comparing to Migkas $A(\ell)$ requires an overall sky-axis orientation: an unknown angle between the CoE–observer axis and Galactic $\ell = 0$. That orientation is absorbed into a phase $\ell = \varphi + \ell_0$ scanned over $[0, 360^\circ)$. The phase is a geometric nuisance parameter, not a new force law.

4 Observables from null geodesics (pipeline)

4.1 What a finished calculation would do

In a completed LTB treatment one integrates past-directed null geodesics from the off-center observer, computes redshift $z(\hat{n})$ and luminosity distance $d_L(z, \hat{n})$, and defines an effective direction-dependent Hubble rate from the low-z expansion

$$d_L(z, \hat{n}) = \frac{cz}{H_{\rm obs}(\hat{n})} + \mathcal{O}(z^2), \tag{10}$$

or, more invariantly, uses the multipole decomposition of the general d_L Hubble law [Heinesen, 2021]. CMB temperature along each ray follows from the redshift of the last-scattering surface (or an equivalent photon distribution), as in Alnes and Amarzguioui [2006], Kodama et al. [2010], Saito et al. [2010].

4.2 What this toy does

The present shot replaces the geodesic d_L map by the Chapter 1 sheet convention applied to the FLRW-shell field:

$$H_{\rm obs}(\varphi) \approx H\big(r(\varphi)\big), \qquad A(\varphi) = \frac{H_{\rm obs}(\varphi)}{\langle H_{\rm obs}\rangle}. \tag{11}$$

This inherits the Newtonian scaffolding: it is honest about being a local-H / sheet-AF proxy, not Etherington reciprocity through curved null congruences.

Amp$\times$ sin and 2ℓ as diagnostic residuals. As in Chapter 1, optional layers

$$A \leftarrow A + \mathrm{Amp}\,\sin(\ell + \phi), \tag{12}$$

$$A \leftarrow \big(1 + \varepsilon\sin(2\ell + \psi)\big)A \tag{13}$$

may be fit after the geometric $A(\ell)$. They are **not** free parameters of the LTB metric. They diagnose leftover phase and quadrupole structure (axis misalignment, non-radial motion, strip geometry of Migkas bins, etc.). The GR claim, such as it is, stops at the metric + off-center geometry + observable map.

5 Numerical toy calculation

5.1 Setup

Parameters (dimensionless, then converted for CMB with $D \sim c/H_0$):

- Mass profile (5) with $D/L = 6$.

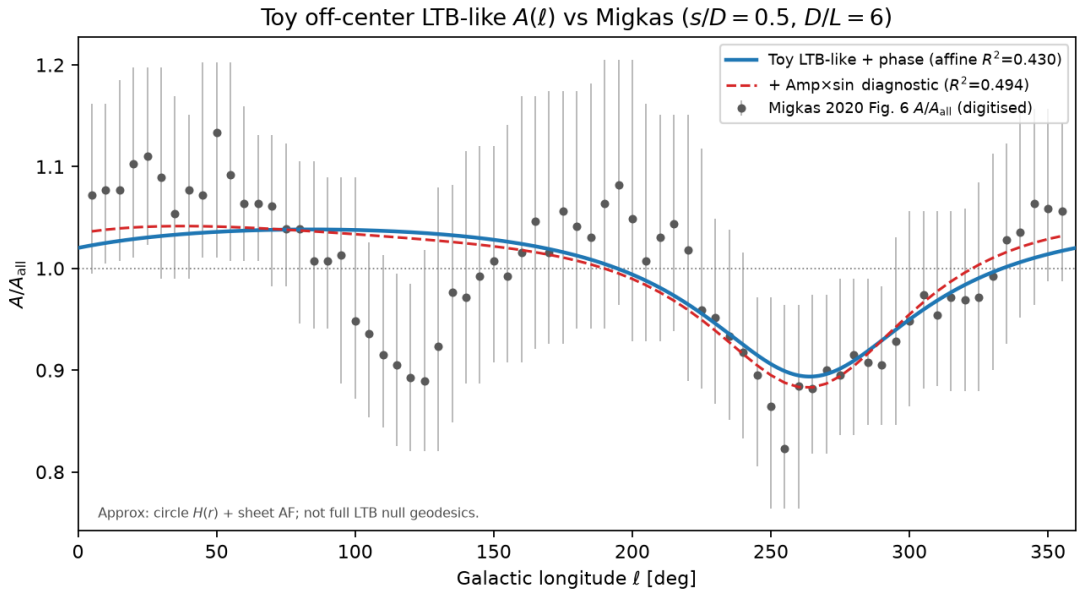


Figure 2: Toy off-center LTB-like $A(\ell)$ (FLRW-shell $H(r)$ + sheet AF, affine-matched after phase) versus digitised Migkas et al. (2020) Fig. 6 $A/A_{\rm all}$. Geometry+phase affine $R^2 \approx 0.430$; adding a diagnostic Amp$\times$ sin layer reaches $R^2 \approx 0.494$. These R^2 values are *not* competitive with the Newtonian Chapter 1 ladder ($0.543 \to 0.746 \to 0.875$) and are not dressed up. Approximation: no full LTB null geodesics.

- Observer at $r = D$; sources on a circle with $s/D = 0.5$ (Newtonian d/D ballpark).
- $H(r)$ from (7); A from (11).
- Migkas comparison: digitised Fig. 6 (`/workspace/cosmology/migkas/fig6_digitized.csv`); affine R^2 after best sky-axis phase.
- Code: `scripts/toy_offcenter_ltb.py` with the project `.venv` Python.

5.2 Migkas $A(\ell)$ result

Reported numbers from the toy run:

Quantity	Value
D/L	6
s/D (source circle)	0.5
Best sky phase ℓ_0	84°
Affine R^2 (geometry + phase)	≈ 0.430
R^2 + Amp$\times$ sin (diagnostic)	≈ 0.494
Best diagnostic Amp	-0.15 (grid edge; weak gain)

The geometric angular shape is broadly dipolar in φ after the sheet map; Migkas $A(\ell)$ has additional structure that this FLRW-shell proxy does not capture. Raising R^2 with Amp$\times$ sin helps only modestly here—unlike Chapter 1—which is itself information: the toy GR backbone is not yet carrying the Newtonian template's full angular content.

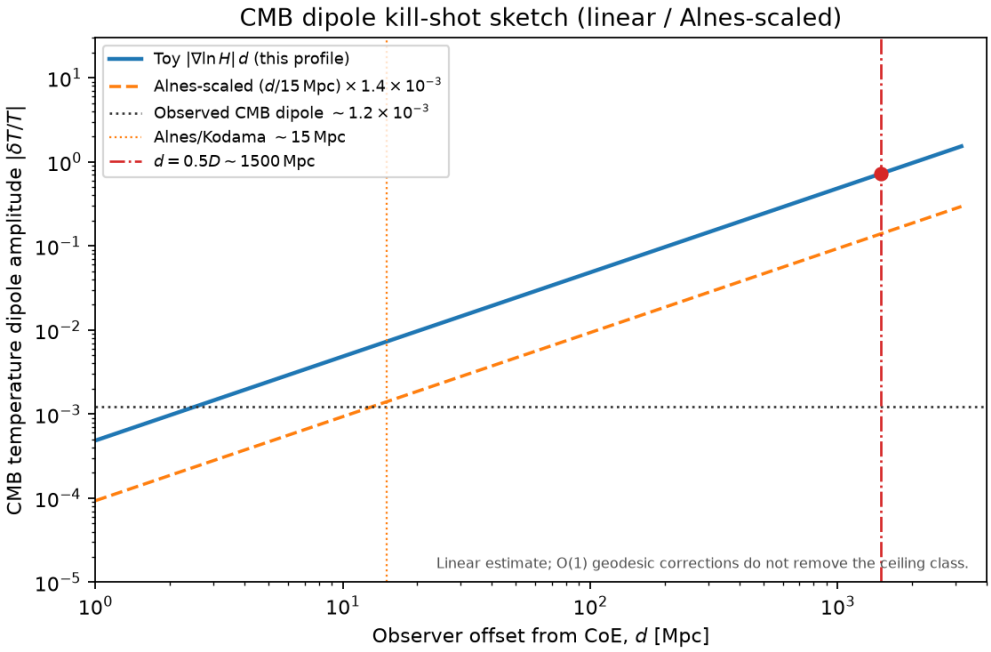


Figure 3: CMB dipole amplitude versus observer offset d. Solid: linear $|\nabla \ln H|\, d$ for the toy profile with $D \sim 3000\,\mathrm{Mpc}$. Dashed: Alnes-scaled sketch $(d/15\,\mathrm{Mpc}) \times 1.4 \times 10^{-3}$. Horizontal: observed CMB dipole $\sim 1.2 \times 10^{-3}$. Vertical: Alnes/Kodama $\sim 15\,\mathrm{Mpc}$ and Newtonian $d = 0.5D$. Full geodesic LTB can change $O(1)$ factors; it does not remove the ceiling class for a horizon-scale spherical offset.

6 CMB kill-shot

Spherical symmetry plus an off-center observer generically produces a large CMB temperature dipole. Alnes and Amarzguioui [2006] showed that LTB void models fitted to SN and the first CMB peak force the observer within $\sim 15\,\mathrm{Mpc}$ of the center once the COBE/WMAP-scale dipole is respected; analytic formulae in Saito et al. [2010], Kodama et al. [2010] recover the same ceiling ($\delta L \lesssim 15$–$16\,\mathrm{Mpc}$). SNIa alone are much weaker on centering [Alnes and Amarzguioui, 2007, Blomqvist and Mörtsell, 2010].

Linear toy estimate. At linear order in offset d,

$$\left|\frac{\delta T}{T}\right|_{\mathrm{dipole}} \sim \left|\nabla \ln H\right| d, \tag{14}$$

with an Alnes-scaled reference curve $(d/15\,\mathrm{Mpc}) \times 1.4 \times 10^{-3}$ matching their Model I a_{10} at 15 Mpc. For the toy profile, evaluating $\nabla \ln H$ at $r = D$ and taking $D \sim 3000\,\mathrm{Mpc}$ (Hubble-length stand-in for a beyond-horizon CoE) gives $|\nabla \ln H| \sim 4.85 \times 10^{-4}\,\mathrm{Mpc}^{-1}$, so the observed dipole $\sim 1.2 \times 10^{-3}$ is reached already at

$$d_{\max} \sim 2.5\,\mathrm{Mpc} \tag{15}$$

for *this* steep centrally concentrated profile—tighter than the void $\sim 15\,\mathrm{Mpc}$ class because the Hubble gradient is larger. At the Newtonian target $d \sim 0.5\,D \sim 1500\,\mathrm{Mpc}$, the same estimate yields $|\delta T/T| \sim 0.7$, orders of magnitude above the data. The Alnes-scaled curve at that offset is ~ 0.14—still fatal.

Why beyond-horizon CoE is in tension. If the CoE lies beyond the observable horizon and we sit at a substantial fraction of the Hubble length from it, spherical symmetry makes our sky look into a huge radial gradient of expansion and gravitational potential. The CMB dipole is the leading symptom. Kinematic subtraction of the observed dipole does not automatically cancel a *geometric* dipole of comparable or larger size without fine-tuned compensation [Alnes and Amarzguioui, 2006].

What would be needed to survive.

1. **Tiny offset:** $d \lesssim \mathcal{O}(10)\,\mathrm{Mpc}$ in spherical models—Copernican fine-tuning relative to a Gpc-scale structure, and incompatible with Newtonian $d/D \sim 0.5$ if D is cosmological.
2. **Break spherical symmetry** enough to suppress the monopole-gradient dipole while retaining a Migkas-like $A(\ell)$—no longer pure LTB; requires controlled anisotropy or screening, with Saadeh et al. Bianchi limits in mind [Saadeh et al., 2016] (see Sec. 7).
3. **Compensating peculiar velocity / ISW** tuned against the geometric dipole at the 10^{-3} level—possible in principle, not natural, and must not ruin other multipoles [Alnes and Amarzguioui, 2006].
4. **Reinterpret the CMB dipole** as mostly cosmological rather than kinematic—a major claim against the standard reading; needs a full Boltzmann+geodesic calculation, not a sketch.

None of these is demonstrated here. The honest verdict of this toy: the Migkas angular program and a beyond-horizon spherical CoE are in direct conflict under standard CMB dipole accounting.

7 Roadmap

Bertschinger (1983). Cosmological self-similar blast waves in Einstein–de Sitter [Bertschinger, 1983] are the closest classical dynamics cousin to the Sedov–Taylor language of Chapter 1. A later upgrade should replace the static toy $M(r)$ with a Bertschinger-like self-similar shell / blast profile inside an expanding background, then recompute off-center null geodesics.

Smoller–Temple (1998, 2002). Shock-wave cosmologies matching FRW to TOV, including placement beyond one Hubble length inside a black hole [Smoller and Temple, 1998, 2003], are the closest GR ontology cousins for a beyond-horizon CoE. They do not by themselves solve the off-center CMB dipole; they upgrade the matching and global causal structure once the local LTB sketch is under control.

Heinesen multipoles. Forecasts for $d_L(z, \hat{n})$ should eventually be phrased in the model-independent multipole Hubble law [Heinesen, 2021], so that Migkas-like directional D_L constraints can be compared without committing to FLRW background language.

What we are not doing: Bianchi. Globally homogeneous but anisotropic Bianchi models are tightly constrained by Planck temperature and polarization [Saadeh et al., 2016]: anisotropic expansion odds $\sim$ 121,000 : 1 against among that class. This program is **inhomogeneous** (spherical CoE + off-center observer), not Bianchi. Saadeh does not directly kill LTB/ST-off-center, but it forbids casually rebranding the anisotropy as universe-wide shear. Any survival route that breaks spherical symmetry must still respect those CMB limits.

8 Conclusions

What this shot achieved.

- A self-contained GR-native note stating LTB + decreasing-$M(r)$ + off-center geometry + sheet-A pipeline.
- A reproducible toy with two required figures: $A(\ell)$ vs Migkas and CMB dipole vs d.
- Real (not fabricated) Migkas comparison: affine $R^2 \approx 0.43$ (geometry+phase); Amp$\times$ sin diagnostic ≈ 0.49.
- An explicit CMB kill-shot ledger aligned with Alnes/Kodama, showing horizon-scale offsets are untenable in spherical symmetry under standard dipole accounting.

What remains open. Full LTB (or LTB+Λ) null geodesics for d_L and $\delta T/T$; Bertschinger self-similar mass profiles; Smoller–Temple global matching; any non-spherical mechanism that could reconcile Migkas-scale $A(\ell)$ with CMB isotropy without Copernican fine-tuning. Until those exist, Chapter 2 is a sketch: metric language for the Newtonian template, and a clear statement of the CMB wall.

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